

Compiler Backend / Code Generation Engineer with professional LLVM 22/C++ experience at MCST. Practical proof points: SSA/CFG validation, liveness/interference, register allocation and spills, phi lowering, SysV x86-64 emission, and native-vs-interpreter differential execution.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 → : Professional LLVM + static analysis

PROFESSIONAL EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer

  - Contribute to SharpChecker, ISP RAS's C#/.NET static-analysis platform.
  - Current work includes symbolic-computation tasks in C#/.NET program analysis.
- July — August 2026

MCST — Compiler Engineering Intern

  - Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
  - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

PROJECTS

- x86-64 Compiler Backend & Register Allocation

SSA/CFG validation → liveness/interference → register allocation/spills → phi lowering → SysV x86-64 emission.

Assignment verifier plus native-vs-interpreter differential execution; the current project CI is green; a separate public native-codegen lineage starts with Visk (2023) and Wist4 (2024).
- LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound.

29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior.
- UniversalToolchain

Dependency/conflict resolution, implementation selection, pass ordering, and typed artifact routes before execution.

1,324/1,324 tests; interpreter/CIL parity and configuration checks before runtime.

SKILLS

- Backend / Codegen

x86-64, SysV ABI, stack frames, calling conventions, liveness/interference, register allocation, spills, phi lowering, native execution
- Low-level / Assembly

Iced.Intel code emission, executable-memory JIT experiments; NASM IA-32, CDECL, x87
- LLVM / IR

C++23, LLVM 22/IR, CFG/SSA, loop/interprocedural analysis, IR transformations
- Verification

assignment verifier, LLVM Verifier, structural checks, idempotence, differential execution

EDUCATION

HSE University — Software Engineering

RECOGNITION

- LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.
- UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.
- MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote